

Project Design Phase-II

Technology Stack (Architecture & Stack)

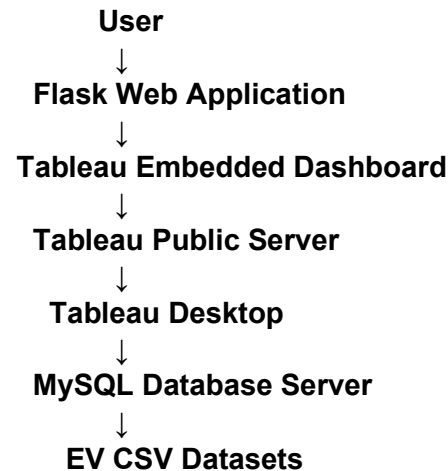
Date	25 June 2026
Team ID	SWTID-2026-4616
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Datasets:

- ElectricCarData
- EVIndia
- Charging Stations
- Cheapest EV Cars

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web page to display EV analytics dashboard	HTML, CSS, JavaScript
2	Application Logic-1	Backend web application integration	Python Flask
3	Application Logic-2	Dashboard creation and data visualization	Tableau Desktop
4	Application Logic-3	Data processing and queries	SQL
5	Database	Stores EV datasets and structured information	MySQL Server 8.0
6	Cloud Database	Published dashboard hosting platform	Tableau Public
7	File Storage	Stores project files and datasets	Local File System, GitHub
8	External API-1	Dashboard embedding service	Tableau Embed API
9	External API-2	Not Applicable	Not Applicable
10	Machine Learning Model	Not Applicable	Not Applicable
11	Infrastructure	Application execution and deployment	Local Server, Flask Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for web integration and development	Flask, Python
2	Security Implementations	Secure database connection and dashboard access management	MySQL Authentication, Tableau Public Access Control
3	Scalable Architecture	Separate database, visualization and web layers	Three-Tier Architecture
4	Availability	Dashboard can be accessed online anytime	Tableau Public
5	Performance	Optimized queries and interactive dashboard loading	SQL Optimization, Tableau Extract

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>